

Homework Studio — Technical Overview

A self-hosted EdTech platform: teaching material in → AI-generated, teacher-approved exams out, plus auto-graded attempts, warm progress reports, and an AI tutor.

Independent portfolio demo. Not affiliated with any company. All data is synthetic. This document explains the architecture, the end-to-end data flow, the database design, and the engineering decisions behind each choice.

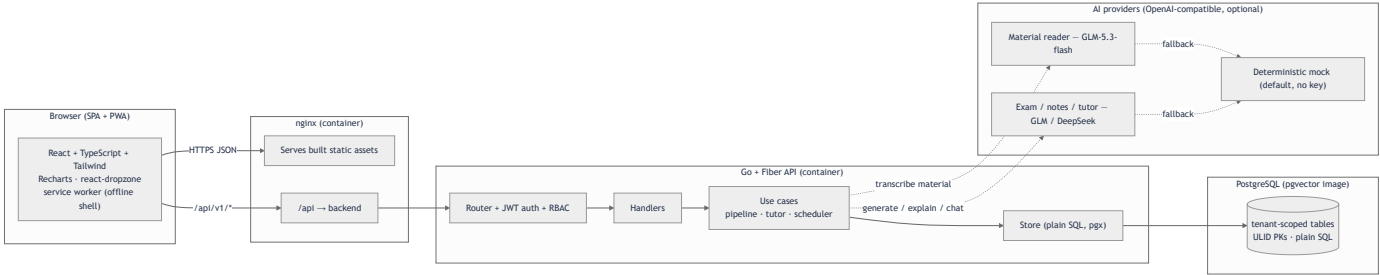
1. What it is

Homework Studio is the *platform layer* around a kids' learning product. It has four roles and one pipeline:

Role	What they do
 Teacher	Upload teaching material (PDF/image/text). The AI reads it and generates a custom exam , teaching notes , and tutor knowledge . Low-confidence questions open a review step so the teacher approves them before publishing.
 Student	A gamified home (XP, levels, streak, badges, leaderboard), take a published exam (auto-graded), practise from the bank (earns XP, not graded), review past attempts, get " Show me how " explanations, and chat with an AI tutor grounded in the teacher's material.
 Parent	A family dashboard — each child's grades <i>and</i> engagement (streak, level, badges), an AI " how to help " tip, compare-to-class , a " what's new " activity feed, and a printable progress report .
 Admin	A school analytics dashboard (mastery bands, at-risk early-warning, roster), an Insights page (engagement/XP, a 14-day trend, and per-exam analytics incl. the hardest question), and the automation controls (a live activity feed of what the background jobs did).

The product's spine is a **document-ingestion → generation pipeline with a human-in-the-loop confidence gate** — the teacher's material is read and turned into an assessment, and the AI's least-confident questions wait for the teacher's approval before students ever see them.

2. Architecture at a glance



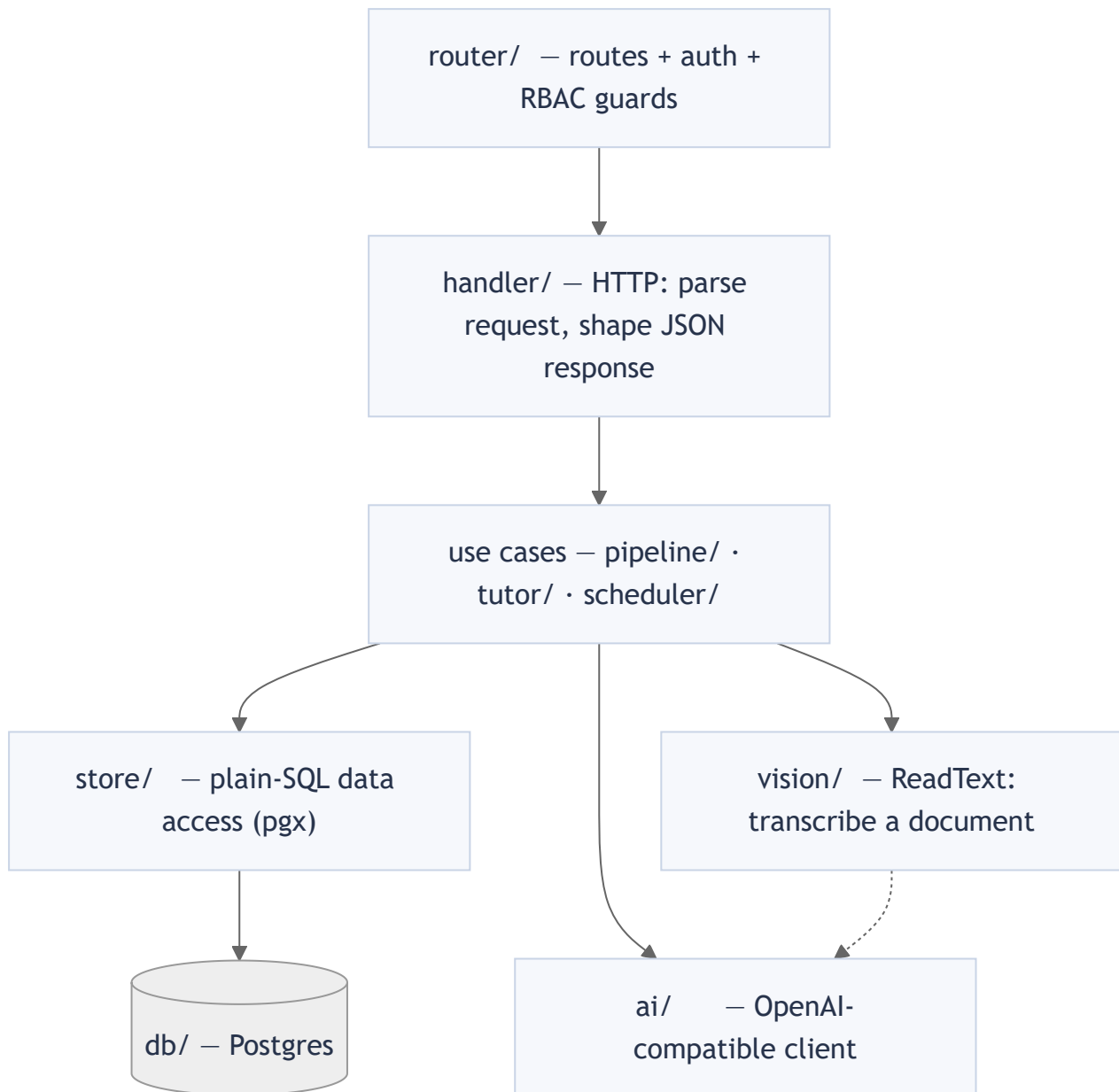
One request path, one data store, swappable AI. The SPA talks only to the Go JSON API. Every AI touchpoint has a deterministic **mock** fallback, so the whole system runs with **no API key** and upgrades to real models with an env change.

3. Tech stack — and *why*

Layer	Choice	Why this, not the obvious alternative
Language	Go 1.26	One static binary, fast cold start, trivial to self-host on a single server. Strong concurrency (the ingest pipeline runs in a goroutine) without extra infrastructure.
HTTP	Fiber v2 (v2.52)	Minimal, express-style router on top of fasthttp. Middleware groups map cleanly onto role guards. Small surface, no framework lock-in.
DB access	pgx v5 + plain SQL (no ORM)	The queries here are analytics-shaped (aggregations, distributions, joins). Hand-written SQL is faster to reason about and tune than an ORM's generated queries, and there is no hidden N+1. <code>pgxpool</code> gives connection pooling.
Migrations	Embedded <code>.sql</code> , applied at startup	No migration CLI to install; the binary owns its schema. <code>CREATE TABLE IF NOT EXISTS</code> keeps startup idempotent.
IDs	ULID (<code>oklog/ulid</code>), generated in Go	Sortable by creation time, URL-safe, no DB round-trip to mint an ID, no sequence contention. Stored as <code>VARCHAR(26)</code> .
Auth	JWT (<code>golang-jwt v5</code>) + <code>bcrypt</code> (<code>x/crypto</code>)	Stateless tokens (no server session store) fit a self-hosted single binary. <code>bcrypt</code> for password hashing.
Database	PostgreSQL (<code>pgvector/pgvector:pg16</code>)	Self-hosted, battle-tested. The image ships <code>pgvector</code> , so the RAG embeddings can move from JSONB to a real vector index without changing databases. Supabase-compatible — Supabase is managed Postgres.
Frontend	Vite + React 18 + TypeScript + Tailwind	Fast dev server + tiny production bundle. A plain SPA (no SSR) because this is an authenticated app, not a content site. The same React/Tailwind components move to Next.js unchanged if that stack is preferred.
Charts / UX	Recharts · react-dropzone · react-markdown + KaTeX	Recharts for the score distributions and trends; react-dropzone for the upload; react-markdown + KaTeX to render the tutor's LaTeX explanations.
PWA	vite-plugin-pwa (Workbox)	Installable app + offline app shell via an auto-updating service worker.
AI client	One OpenAI-compatible client	GLM, DeepSeek, TypeAI (jev), and OpenAI all speak the same <code>/chat/completions</code> shape — only base URL / key / model change. No per-vendor SDKs.
Deploy	Docker Compose	<code>postgres + backend + frontend</code> on one host behind nginx. One <code>docker compose up</code> .

4. Clean architecture — how the code is layered

Business logic (grading, gating, explanations, practice generation) is isolated from HTTP and SQL, so it is unit-testable and provider-agnostic.



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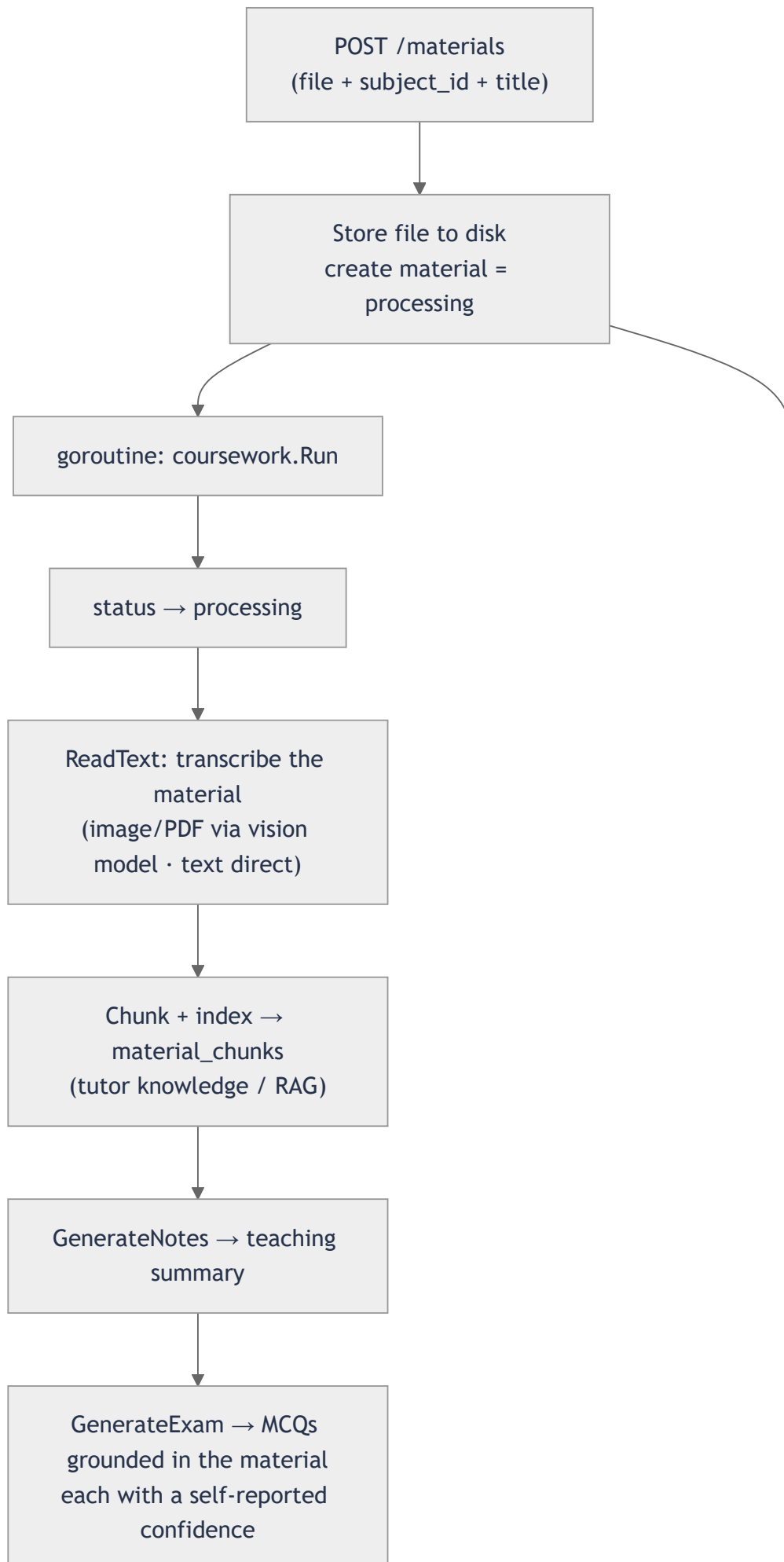
backend/
├── cmd/
│   ├── api/      server entrypoint (wires config → store → coursework/tutor/scheduler → router)
│   └── seed/     one-shot demo seed (1 school, 4 logins, 2 subjects, bank, exams, attempts)
└── internal/
    ├── config/   env loading + provider switches
    ├── db/       connection pool + embedded migrations (applied at startup)
    ├── domain/   entities + enums (Material, Exam, Question, ...) + ULID mint
    ├── auth/     JWT issue/verify, bcrypt
    ├── middleware/ RequireAuth, RequireRole
    ├── store/    plain-SQL access – one file per aggregate (material, exams, tutor, admin, ...)
    ├── vision/   ReadText: transcribe a document (image/PDF via the vision model, text direct)
    ├── ai/       tiny OpenAI-compatible chat client (GLM/DeepSeek/TypeAI/OpenAI) + ChatVision
    ├── coursework/ read material → index (RAG) → teaching notes → generate exam → gate
    ├── tutor/    exam generator, teaching notes, explanations + grading, RAG chat
    ├── scheduler/ background job: weekly reports + recap data + auto-flagging
    ├── handler/  HTTP handlers
    └── router/   route table

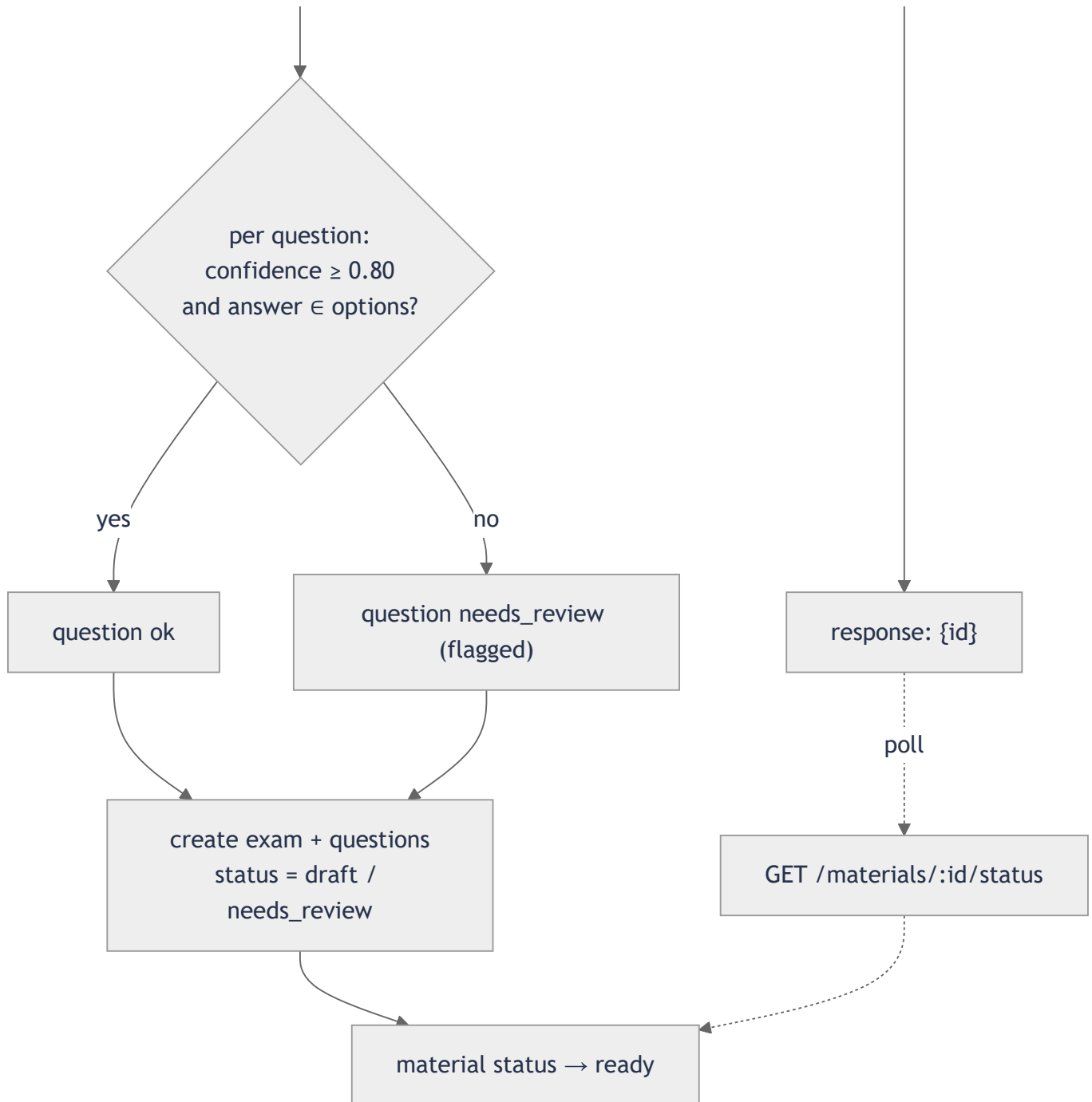
```

Dependency rule: handlers depend on use cases; use cases depend on the store and on the `ai.Client` *interface* — never on a concrete provider. Swapping GLM for DeepSeek, or a real model for the mock, is a constructor argument (or an env flag), not a code change.

5. The coursework pipeline — material → exam

The heart of the ingest → generation side. A teacher uploads teaching material; the handler stores the file, creates a `processing` material, and kicks the pipeline off **asynchronously** (in a goroutine) so the request returns immediately. The frontend polls status.





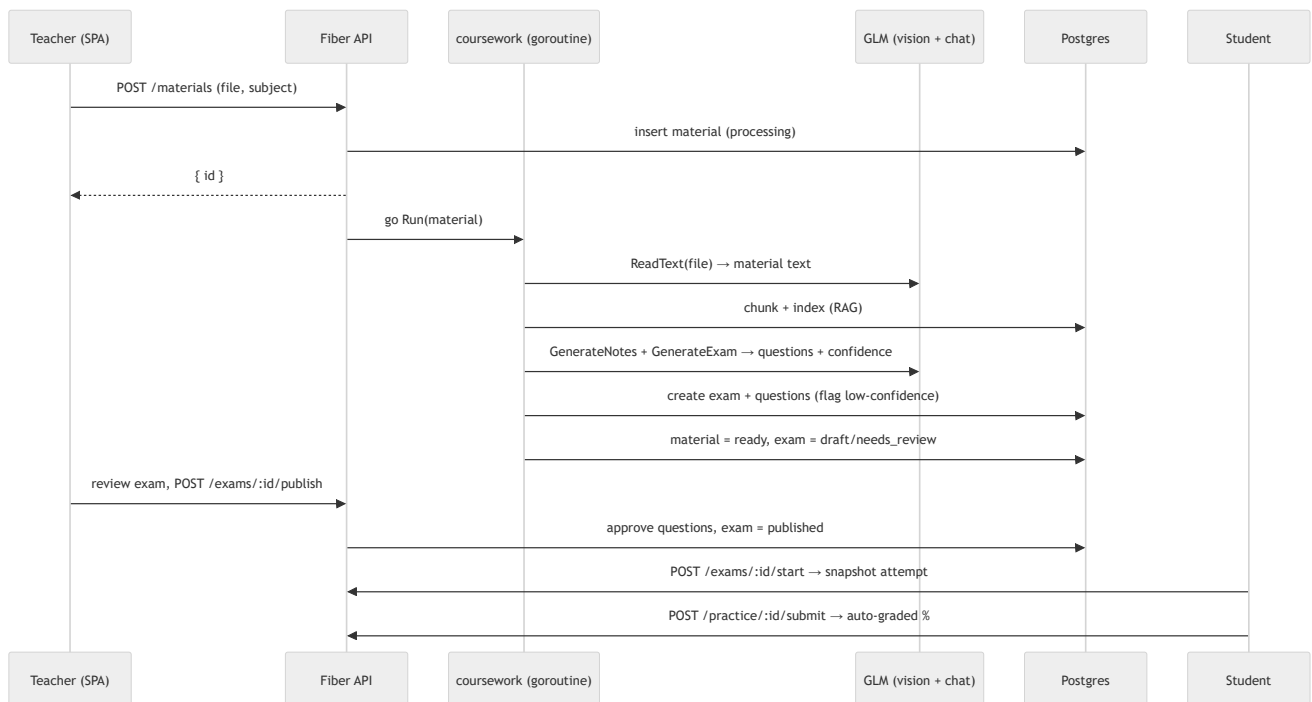
The confidence gate — now on the *generated* questions:

generate questions (per-question confidence) $\rightarrow$ GATE $\rightarrow$ decide

every question $\geq$ threshold	$\longrightarrow$	draft (ready to publish)
any question $<$ threshold	$\longrightarrow$	needs_review $\rightarrow$ teacher approves

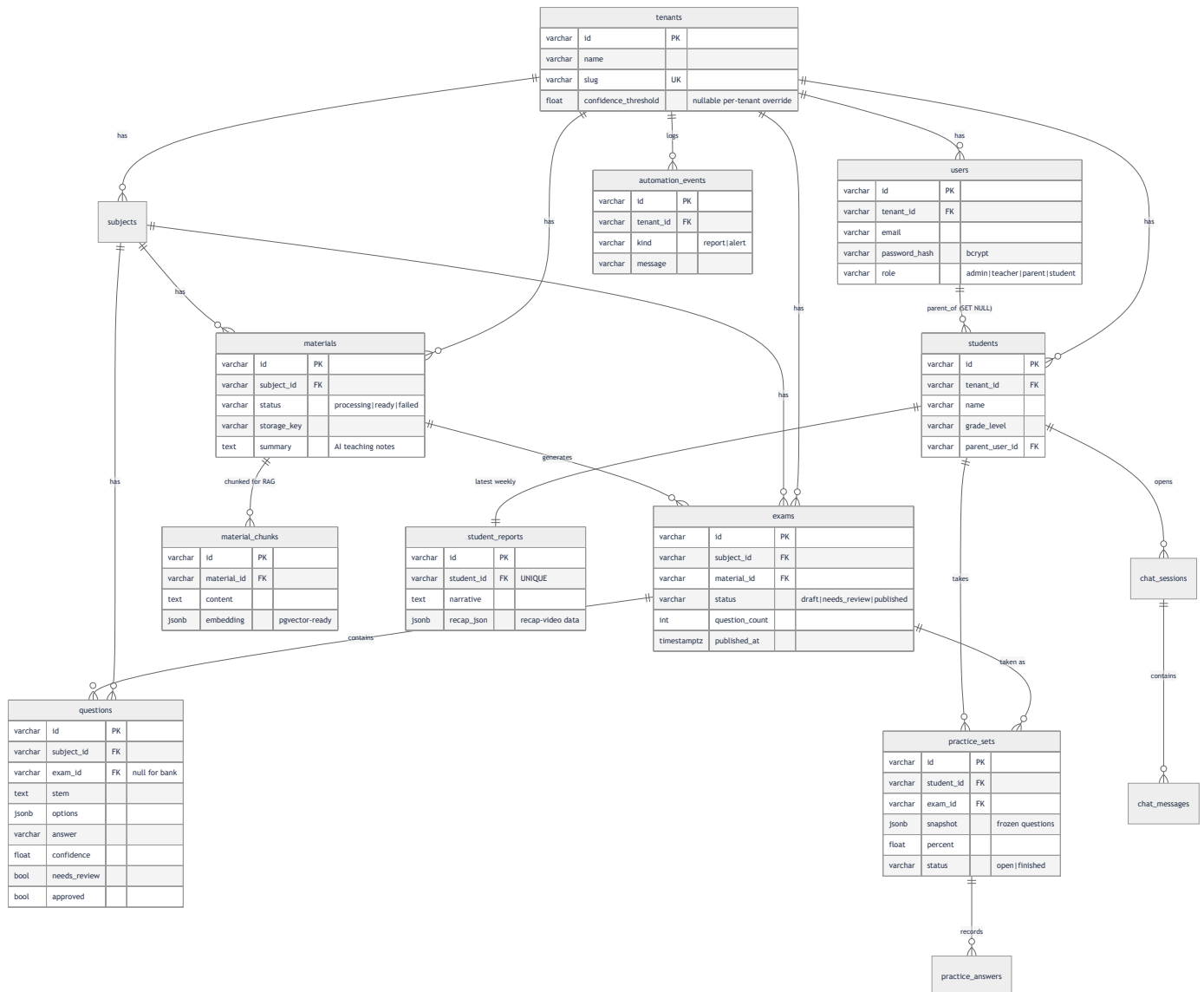
An exam is **published only after the teacher approves it**. Questions the model was least sure about are **flagged**; the teacher reviews them (keep, or discard), then publishes. This is human-in-the-loop by construction: the AI never puts a question in front of students that the teacher hasn't signed off on.

Sequence: upload → generate → review → publish → take



6. Database design (ERD)

Plain Postgres, **ULID string PKs generated in Go**, every business table scoped by `tenant_id` for multi-school isolation. Schema is applied at startup from embedded SQL (`0001_init.sql` ... `0004_coursework.sql`).



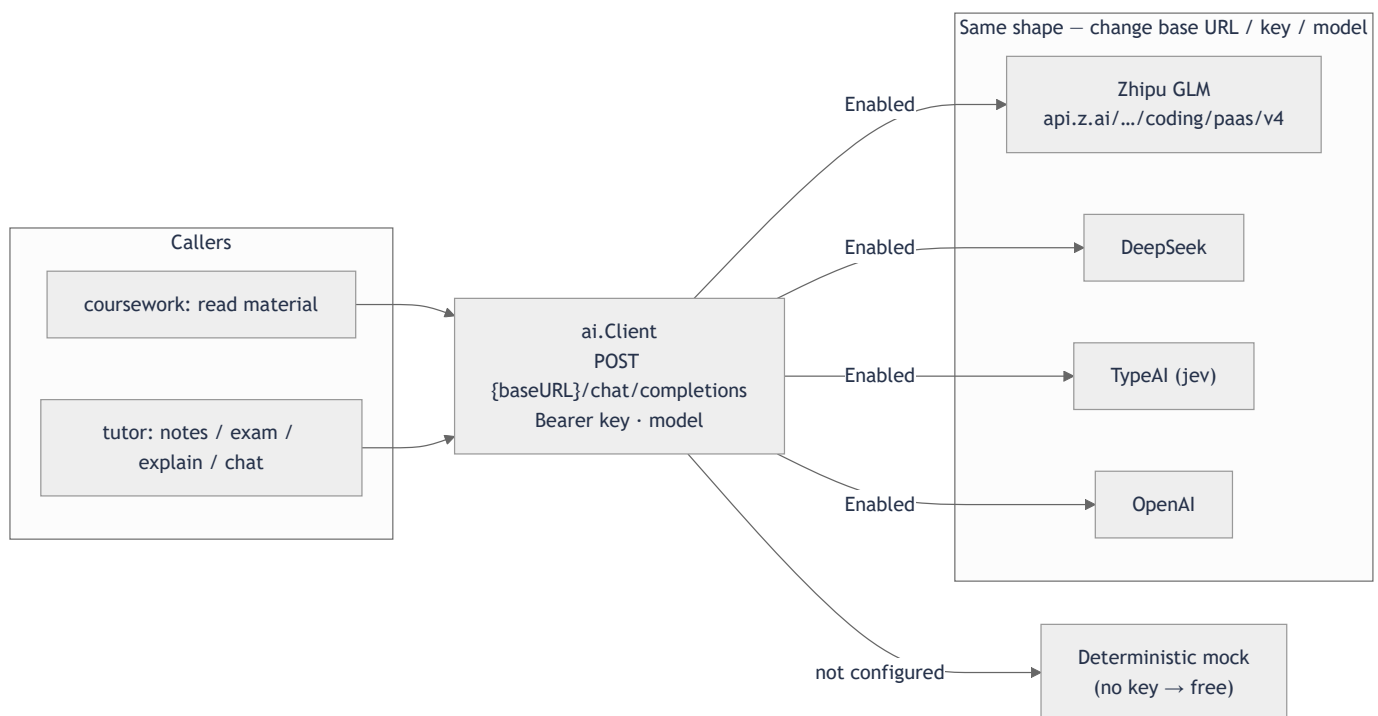
Notes on the design

- questions.confidence + needs_review + approved** are the gate: a generated question below the threshold (or whose answer isn't among its options) is flagged and un-approved until the teacher publishes. Bank questions keep the defaults (confidence 1, approved) and carry a null `exam_id`.
- exams** groups generated questions; `status` walks `draft / needs_review → published`. A student attempt is a **practice_sets** row linked by `exam_id`.
- practice_sets.snapshot** freezes the exact questions (JSONB) when the attempt starts, so it grades against what the student saw — a resumable, tamper-resistant attempt. Finished attempts (`status='finished'`, `percent`) are the single source for all progress/dashboard/admin analytics.
- Student **progress, class stats, and the admin roster all derive from finished practice_sets** — there is no separate results table. Grades count only exam attempts (`exam_id IS NOT NULL`); ungraded practice (`exam_id IS NULL`) still earns XP.
- Gamification (XP / level / streak / badges) and the leaderboard are computed on the fly** from those attempts — no points ledger or badge tables. Per-question answers are saved to **practice_answers** so an attempt can be reviewed.
- material_chunks.embedding** is JSONB today (cosine similarity in Go). The Postgres image already ships **pgvector**, so this becomes a real `vector` column
 - ANN index with no database migration.

- `student_reports` has a **UNIQUE** `student_id` : the scheduler *upserts* one latest report per student. `recap_json` carries the data a recap video renders from.
- `ai_usage` (not drawn) logs prompt/completion tokens per feature per tenant — the metering hook for a real deployment.

7. AI architecture — swappable, mock by default

Everything AI goes through one **OpenAI-compatible** client. `Enabled()` is true only when both a key and base URL are set; otherwise callers fall back to a deterministic mock, so the demo is free and reproducible.



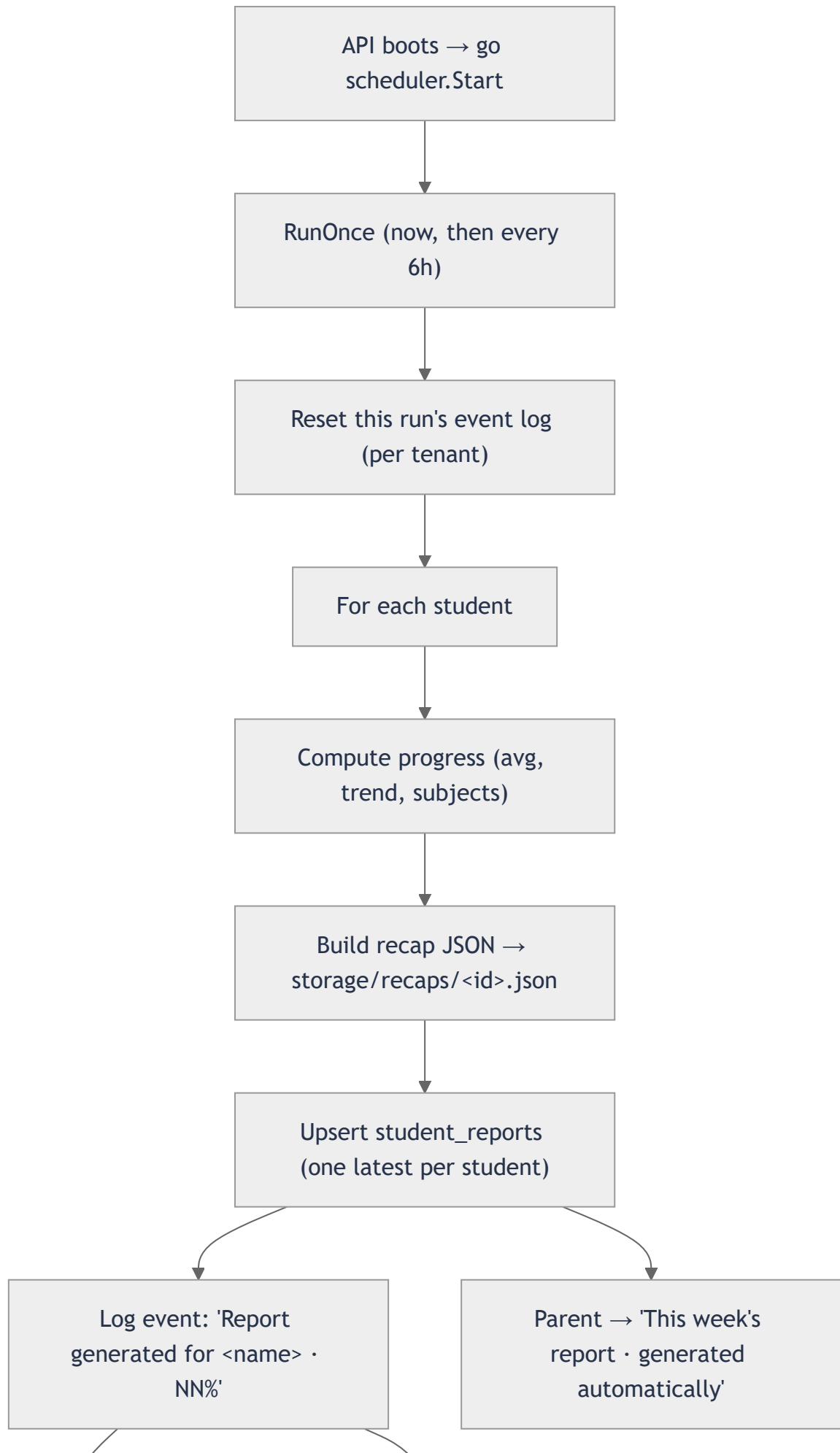
Touchpoint	Env	Real provider	Fallback
Material reader (vision)	<code>VISION_PROVIDER=glm</code> , <code>VISION_MODEL=glm-5.3-flash</code>	GLM transcribes the uploaded PDF/image to text (PDFs rasterized with poppler <code>pdftoppm</code> first).	Text files read directly; a labelled placeholder otherwise.
Exam + notes + tutor	<code>AI_PROVIDER=glm</code> , <code>AI_MODEL=glm-5.3</code> , <code>AI_BASE_URL=.../coding/paas/v4</code>	GLM authors exam questions (with confidence) grounded in the material, writes teaching notes + LaTeX explanations, and answers RAG chat.	Bank-sampled exam / excerpt notes / mock explanation / canned chat.

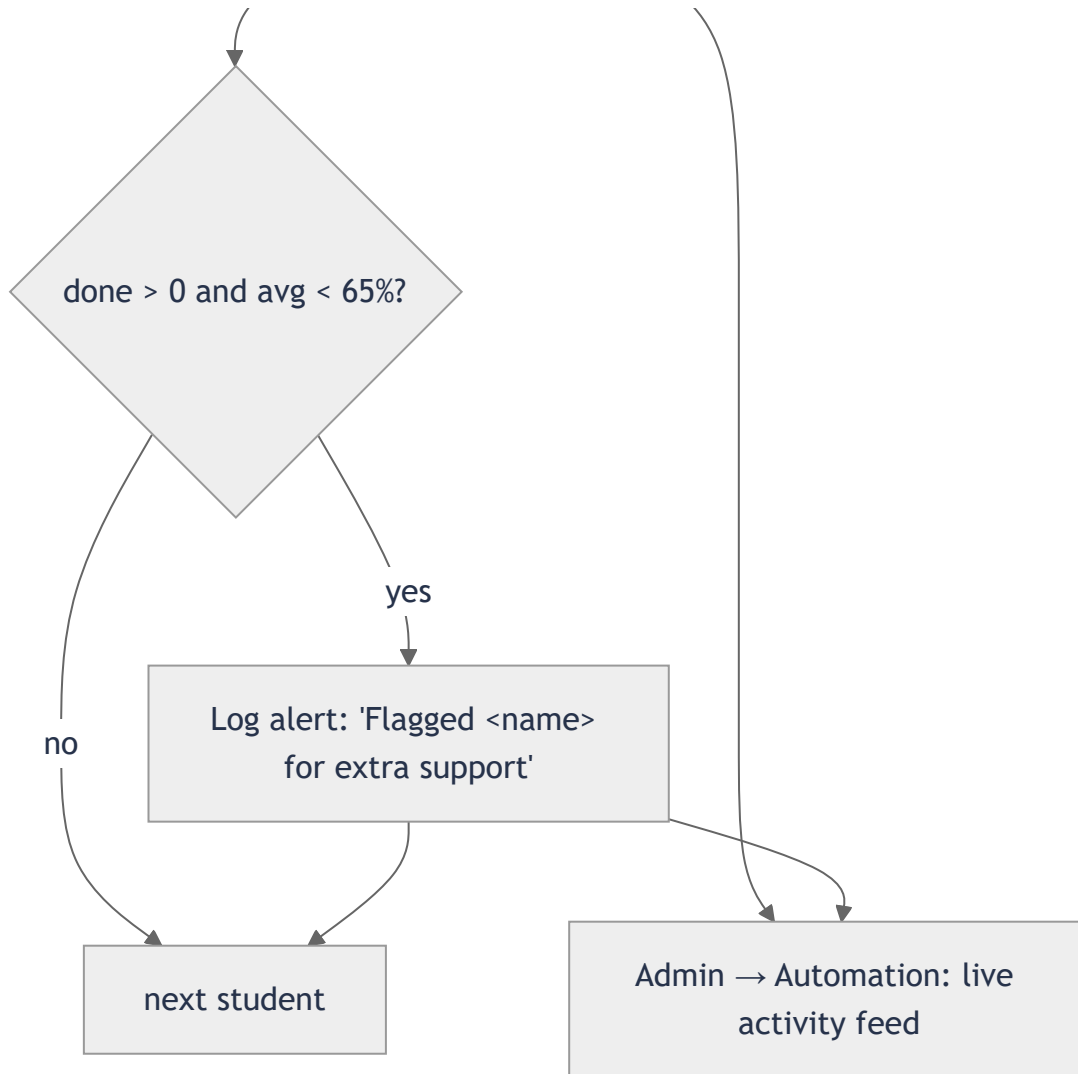
The client is OpenAI-compatible, so **DeepSeek**, **TypeAI (jev)**, or **OpenAI** drop in by changing base URL + key + model. The **confidence threshold** for flagging a generated question is `0.80` (a question is also flagged if its answer isn't among its options).

8. Automation — the hands-off layer

Two things run without anyone pressing a button:

1. **The coursework pipeline** (§5) — every material read, indexed, and turned into a gated exam on upload.
2. **A background scheduler** — runs once on startup and then every `SCHEDULER_INTERVAL` (default `6h`).





The `automation_events` log is what surfaces the automation in the UI: the **admin Automation page** shows a live feed (reports written + students auto-flagged) with a **Run now** trigger, and the **parent page** shows the auto-generated report card. That is how a background job becomes something a user can see.

9. API surface

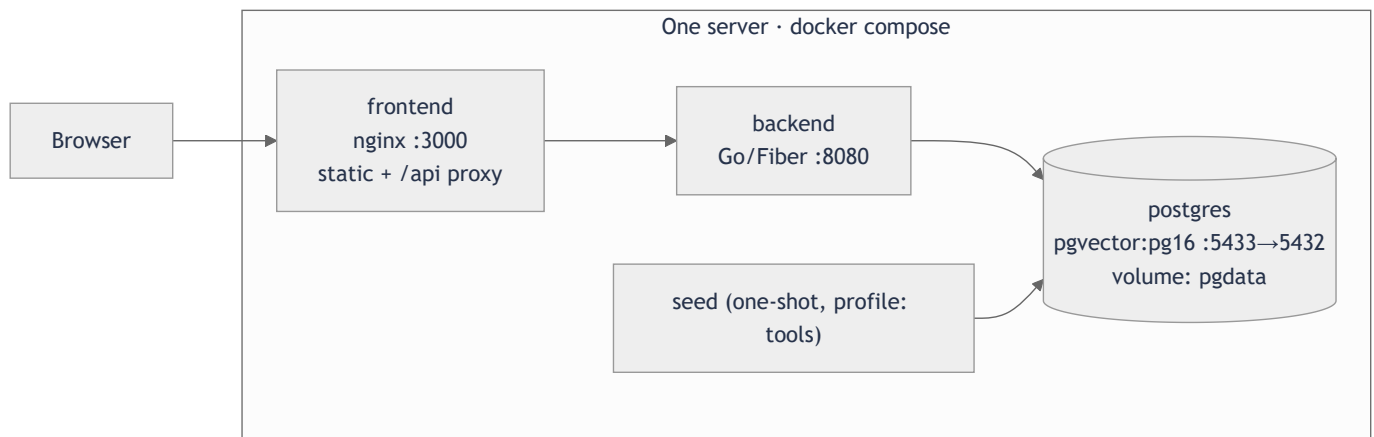
All under `/api/v1`. Auth is a Bearer JWT; roles are enforced by middleware.

Method	Path	Role	Purpose
POST	/auth/login	public	Email + password → JWT
GET	/auth/me	any	Current user
GET	/subjects	any	Subjects for the tenant
GET	/students	any	Students (scoped)
GET	/students/:id/progress	any	Average, timeline, subject strengths
GET	/reports/student/:id	any	Latest auto-generated weekly report
GET	/family · /family/activity	parent	Children (grades + engagement) / "what's new" feed
GET	/students/:id/vs-class · /students/:id/tip	parent	Child vs class average / AI "how to help" tip
POST	/materials	teacher, admin	Upload material → kicks off the coursework pipeline
GET	/materials · /materials/:id · /:id/status	teacher, admin	List / notes / poll status
GET	/exams · /exams/:id	teacher, admin	List generated exams / one with its questions
POST	/exams/:id/publish	teacher, admin	Approve remaining questions → publish
POST	/exams/:id/questions/:qid/discard	teacher, admin	Drop a rejected question
GET	/dashboard/class	teacher, admin	Class stats + distribution
GET	/admin/overview · /admin/automation	admin	School analytics / scheduler feed
GET	/admin/engagement · /admin/trend · /admin/teaching	admin	Engagement / 14-day trend / per-exam analytics
POST	/admin/automation/run	admin	Trigger the scheduler now
GET	/exams/published	any	Published exams a student can take
POST	/exams/:id/start	any	Snapshot a published exam into an attempt
POST	/practice/generate	any	Ungraded practice set from the bank (earns XP)
POST	/practice/:id/submit	any	Auto-grade an attempt (persists per-question answers)
GET	/students/:id/gamification	any	XP / level / streak / badges (derived)
GET	/students/:id/attempts · /attempts/:id/review	any	Results list / review one attempt
GET	/leaderboard	any	Class ranking by XP

Method	Path	Role	Purpose
GET	/questions/:id/explain	any	Step-by-step explanation
POST	/tutor/chat	any	RAG tutor chat

Auth & RBAC. `RequireAuth` verifies the JWT and loads the user; `RequireRole` guards teacher/admin and admin-only groups. Every query is scoped by `tenant_id`, and ownership checks stop a parent from reading another child's data (returns 403).

10. Deployment



```
# 1. Start Postgres + backend + frontend
docker compose -f docker/docker-compose.yml --profile full up -d --build
# 2. Seed the demo dataset
docker compose -f docker/docker-compose.yml --profile tools run --rm seed
# 3. Open → frontend http://localhost:3000 · API http://localhost:8080/health
```

Config is environment-first (`internal/config`). Real AI is opt-in: copy `docker/.env.example` → `docker/.env` (gitignored) with a GLM key and `docker compose up` runs on real models. Defaults keep the stack on the free mock.

Variable	Default	Meaning
<code>DATABASE_URL</code>	local Postgres	pgx connection string
<code>AI_PROVIDER</code> / <code>AI_MODEL</code> / <code>AI_BASE_URL</code> / <code>AI_API_KEY</code>	mock	Exam + notes + tutor
<code>VISION_PROVIDER</code> / <code>VISION_MODEL</code>	mock / <code>glm-5.3-flash</code>	Material reader (transcription)
<code>SCHEDULER_ENABLED</code> / <code>SCHEDULER_INTERVAL</code>	true / 6h	Background reports
<code>ACCESS_TOKEN_TTL_MINUTES</code>	720	JWT lifetime

11. Design decisions & trade-offs

- **No ORM.** The read side is analytics (aggregations, distributions). Plain SQL is clearer and faster to tune here, and there is no hidden query behavior. The cost — writing SQL by hand — is small at this schema size.
- **Async pipeline, poll for status.** Upload returns immediately and the coursework pipeline runs in a goroutine; the SPA polls `/materials/:id/status`. Simple and dependency-free (no queue broker). For higher volume this is where a real worker/queue slots in — the pipeline is already a self-contained unit.
- **Mock-by-default AI.** The demo is free and reproducible; every AI path degrades gracefully to a deterministic result. Real models are one env flag away.
- **JSONB embeddings, pgvector image.** Keeps the demo dependency-light while leaving a zero-migration path to a real vector index.
- **Stateless JWT.** No session store to run; fits a single self-hosted binary.
- **SPA on a JSON API.** The frontend is decoupled; the identical React/Tailwind components port to Next.js unchanged if server rendering is later wanted.

12. Verify it works

```
cd backend && go build ./... && go vet ./...    # compiles clean
cd frontend && npm run build                    # tsc --noEmit + vite build
```

The end-to-end flow: teacher uploads material → the AI generates an exam and flags a low-confidence question → teacher reviews and publishes → a student takes the exam, is auto-graded, and gets an explanation → a parent sees the progress → the admin watches the scheduler log reports and at-risk flags.

See [MANUAL.md](#) for the per-role user guide, and [case-study.md](#) for how each piece maps to a real product's needs.